

SAM TAYLOR

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EDUCATION

Georgia Institute of Technology

Online Master of Science in Analytics

January 2026 - December 2027

GPA: 3.7

Virginia Commonwealth University

Bachelor of Science in Computer Science

January 2021 - December 2023

GPA: 3.5

PROJECTS

Predicting Solar Wind Conditions at Psyche 16 with NASA | Machine Learning, Data Visualization

- Led data collection, cleaning, and exploration to construct usable machine learning datasets.
- Applied time series and regression machine learning models to describe space weather.
- Evaluated model performance using holdout testing to assess predictive reliability.
- Developed visualizations to communicate trends, model behavior, and prediction uncertainty.

Protein Sequence Classifier | Pandas, Random Forest, SVM

- Performed data quality checks for missing and duplicate observations before training.
- Used Random Forest feature importance to reduce dimensionality.
- Trained and evaluated One-vs-Rest SVM classifier to classify sequences using a training/testing split.
- Adjusted SVM threshold to increase detection of minority class, encouraging further investigation.

Capital Bikeshare Usage Analysis | R, Excel, CUSUM, ARIMA

- Cleaned and analyzed over one year of Capital Bikeshare trip data, standardizing timestamps.
- Built Excel pivot tables and dashboards to analyze ridership by hour, weekday, and station location.
- Applied CUSUM to identify changes in ridership patterns and ARIMA modeling to forecast trends.
- Identified peak commuting periods and the most frequently used stations.

EXPERIENCE

Junior Developer, MicroHealth LLC, Vienna, Virginia

February 2024 - September 2024

- Implemented K-Nearest Neighbors (KNN) to power college recommendation search feature, matching high school students with colleges based on their preferences.
- Containerized and integrated Hugging Face dataset using Docker to safeguard PII.
- Engineered multiple AI assistants using prompt engineering to enable scalable AI models.
- Collaborated in full-stack debugging with TypeScript, C# and more, improving system stability.

PROFESSIONAL DEVELOPMENT

Independent Technical Study

October 2024 - August 2025

- Earned Meta Front-End Developer and DeepLearning.AI Agentic AI certifications.
- Completed hands-on training in Docker, AWS Lambda, and RDS; built and deployed portfolio.

George Mason University

Graduate Certificate in Data Analytics

August 2025 - December 2025

GPA: 3.8

TECHNICAL SKILLS

Machine Learning / Data Science: Linear Regression, SVM, Clustering, Forecasting, Decision Trees

Programming Languages: Python, R, SQL, Java, Typescript, C#

Tools: Tableau, Power BI, Excel, MS Office Suite, MySQL, PostgreSQL

ML Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch

Cloud / DevOps: AWS (S3, Lambda, EC2, RDS), Docker, CI/CD, Git, Containerization